

# Nonlinear Systems Overview

Leding Ren

March 31, 2026



# Contents

|                                                               |           |
|---------------------------------------------------------------|-----------|
| <b>Preface</b>                                                | <b>1</b>  |
| <b>Introduction</b>                                           | <b>3</b>  |
| <b>1 Where the Problem Begins: Usability Before Exactness</b> | <b>5</b>  |
| 1.1 Existence vs Usefulness . . . . .                         | 5         |
| 1.2 Limits of Exact Dynamics . . . . .                        | 5         |
| 1.3 Usability as a Primary Constraint . . . . .               | 6         |
| 1.4 Restatement . . . . .                                     | 6         |
| <b>2 Noise as the Boundary of Operational Systems</b>         | <b>7</b>  |
| 2.1 Noise as a Structural Boundary . . . . .                  | 7         |
| 2.2 Uncontrollable and Unobservable Subspaces . . . . .       | 7         |
| 2.3 Internalization and State Expansion . . . . .             | 8         |
| 2.4 Determinism and Randomness . . . . .                      | 8         |
| 2.5 Why Noise Cannot Be Eliminated . . . . .                  | 8         |
| 2.6 Restatement . . . . .                                     | 8         |
| <b>3 Dissipation as a Structural Condition</b>                | <b>11</b> |
| 3.1 Degeneracy in Non-Dissipative Systems . . . . .           | 11        |
| 3.2 Basins and Structural Robustness . . . . .                | 11        |
| 3.3 Limits of Linear Systems . . . . .                        | 12        |
| 3.4 Dissipation as a Necessary Condition . . . . .            | 12        |
| 3.5 Destruction Before Structure . . . . .                    | 12        |
| 3.6 Dissipation and the Arrow of Time . . . . .               | 13        |
| 3.7 Restatement . . . . .                                     | 13        |
| <b>4 Randomness and Dimension</b>                             | <b>15</b> |
| 4.1 Representation and Collapse . . . . .                     | 15        |
| 4.2 Structural Limits of Finite Description . . . . .         | 15        |
| 4.3 Accumulation and Reorganization . . . . .                 | 16        |
| 4.4 Randomness as a Structural Consequence . . . . .          | 16        |
| 4.5 Determinism, Chaos, and Operational Equivalence . . . . . | 16        |
| 4.6 Dimension as Representational Capacity . . . . .          | 17        |
| 4.7 Restatement . . . . .                                     | 17        |
| <b>5 Trapping Regions and the Limits of Reversibility</b>     | <b>19</b> |

|          |                                                                |           |
|----------|----------------------------------------------------------------|-----------|
| 5.1      | Reversible Dynamics and Empty Motion . . . . .                 | 19        |
| 5.2      | A Thought Experiment . . . . .                                 | 20        |
| 5.3      | Measure Preservation and the Absence of Contraction . . . . .  | 20        |
| 5.4      | Recurrence Without Attraction . . . . .                        | 21        |
| 5.5      | Basins and Observable Structure . . . . .                      | 21        |
| 5.6      | Trapping Regions and the Emergence of Structure . . . . .      | 21        |
| 5.7      | Reversible Systems Admit No Trapping Region . . . . .          | 22        |
| 5.8      | Observable Structure Requires Trapping Regions . . . . .       | 22        |
| 5.9      | Exact Existence and Observable Meaning . . . . .               | 23        |
| <b>6</b> | <b>Information, Records, and Irreversibility</b>               | <b>25</b> |
| 6.1      | Irreversibility Creates Records . . . . .                      | 25        |
| 6.2      | Records Compress History . . . . .                             | 25        |
| 6.3      | Observation as Record Interpretation . . . . .                 | 26        |
| 6.4      | Maxwell's Demon and the Cost of Records . . . . .              | 26        |
| 6.5      | Information, Entropy, and Control . . . . .                    | 26        |
| 6.6      | From Records to History . . . . .                              | 27        |
| <b>7</b> | <b>Thermodynamics</b>                                          | <b>29</b> |
| 7.1      | Irreversibility and Trajectory Elimination . . . . .           | 29        |
| 7.2      | Trajectory Compression . . . . .                               | 29        |
| 7.3      | Trapping Regions . . . . .                                     | 30        |
| 7.4      | Closure and Structure . . . . .                                | 30        |
| 7.5      | Records and Compressed History . . . . .                       | 31        |
| 7.6      | Thermodynamics as Structural Description . . . . .             | 31        |
| 7.7      | Thermodynamic Laws as Structural Constraints . . . . .         | 31        |
| <b>8</b> | <b>Limits of Representation</b>                                | <b>35</b> |
| 8.1      | The Boundary of Observation . . . . .                          | 36        |
| 8.2      | Finite Propagation and Observational Horizon . . . . .         | 37        |
| 8.3      | General Relativity and the Geometry of Accessibility . . . . . | 37        |
| 8.4      | Distinguishability-Preserving Evolution . . . . .              | 38        |
| 8.5      | Quantized Records and the Formation of Outcomes . . . . .      | 39        |
| 8.6      | Direction Without Time . . . . .                               | 40        |
| 8.7      | Symmetric Branches and Antimatter . . . . .                    | 40        |
| 8.8      | Labels, Interactions, and Force . . . . .                      | 42        |
| 8.9      | Gravity as Geometric Bias . . . . .                            | 42        |
| 8.10     | Undifferentiated Residue and Structural Release . . . . .      | 43        |
| 8.11     | Boundary Remark . . . . .                                      | 44        |

---

|          |                                                    |           |
|----------|----------------------------------------------------|-----------|
| <b>9</b> | <b>Structural Viewpoint</b>                        | <b>47</b> |
| 9.1      | Structural Universality . . . . .                  | 47        |
| 9.2      | Usable and Operational Systems . . . . .           | 47        |
| 9.3      | Structural Impossibility and Compromise . . . . .  | 48        |
| 9.4      | The Infinite-Dimensional Boundary . . . . .        | 49        |
| 9.5      | Observation and Structural Accessibility . . . . . | 49        |
| 9.6      | Four Structural Constraints . . . . .              | 50        |
| 9.7      | Thermodynamics as Structural Language . . . . .    | 50        |
| 9.8      | Structural Compass . . . . .                       | 51        |
|          | <b>Epilogue</b>                                    | <b>53</b> |



# Preface

This book did not begin as a standard introduction to nonlinear systems. It began from a recurring practical difficulty: many systems remain mathematically describable while becoming progressively less usable under finite observation, finite control, and finite representation. The question pursued here is therefore not simply how nonlinear systems can be solved, but what structural conditions repeatedly appear when such systems must remain operational in practice.

The result is not a textbook in the usual sense. It is not a complete theorem–proof treatment of nonlinear dynamics, nor is it an attempt to replace existing physical laws or mathematical models. It is instead a structural framework. Its aim is to identify recurrent constraints, recurrent tradeoffs, and recurrent structural consequences that appear across different domains when exact histories cannot be fully preserved.

The intended reader is therefore not limited to one discipline. This book is written for readers interested in nonlinear systems, control, information, thermodynamics, and the conceptual foundations of structure under finite conditions. Some readers may come from mathematics, some from physics or engineering, and some from systems theory more broadly. What unifies the audience is not a shared technical background in one field, but an interest in whether these recurring phenomena admit a common structural reading.

The arguments in this book are correspondingly mixed in form. Some are informal guiding statements. Some are local formal definitions or propositions. Some are structural interpretations that organize familiar phenomena within a single frame. The mathematical expressions included here are used to compress structural relations, not to provide maximal formalization at every step. Where the claims are stronger, the scope should be read more carefully. Where the scope is broader, the claims should be read more structurally.

One way to read the book is straight through, from the introduction to the epilogue. Another is to read the introduction, the first chapter, the final synthesis, and the epilogue first, then return to the middle chapters as supporting development. The chapters are arranged to support either approach.

The perspective advanced here is limited, but deliberately so. It does not begin with complete solutions. It begins with limits, and asks what kind of structure remains once those limits are taken seriously.





# Introduction

Nonlinear systems appear in many contexts. They arise in physical processes, biological behavior, engineered systems, and everyday situations where small changes may lead to disproportionately large consequences. Their behavior is often irregular, sensitive, and difficult to predict.

These features are commonly interpreted as signs of complexity. Nonlinear systems are therefore often described as systems that are simply too complicated to be understood in a unified way.

This book takes a different starting point.

Across different domains, similar difficulties repeatedly appear when nonlinear systems are analyzed or operated. Exact solutions are rare. Predictions degrade over time. Observations remain incomplete. Control authority is limited. Noise cannot be eliminated. Trade-offs between robustness and accuracy appear unavoidable.

These difficulties are usually treated as technical obstacles specific to individual models. Their recurrence across unrelated fields suggests that they may not be incidental. They may reflect something more fundamental about how systems operate when they must remain usable.

Real systems do not operate with unlimited precision or complete information. They function under constraints. What can be observed is limited. What can be controlled is restricted. What can be represented is necessarily incomplete. Under such conditions, behavior is shaped not by the existence of exact solutions, but by the constraints under which the system must operate.

From this perspective, nonlinearity does not arise from excessive complexity. It arises from constraint.

This work presents a structural overview of such constraints. Rather than seeking a general solution theory for nonlinear equations, it examines the conditions that repeatedly appear when systems must remain operational in practice. The goal is not to remove uncertainty or irreversibility, but to clarify why they cannot be eliminated.

The scope of this work is therefore limited. It does not replace existing physical laws or mathematical models, nor does it attempt to unify them. Instead, it provides a structural viewpoint from which recurring phenomena can be interpreted.

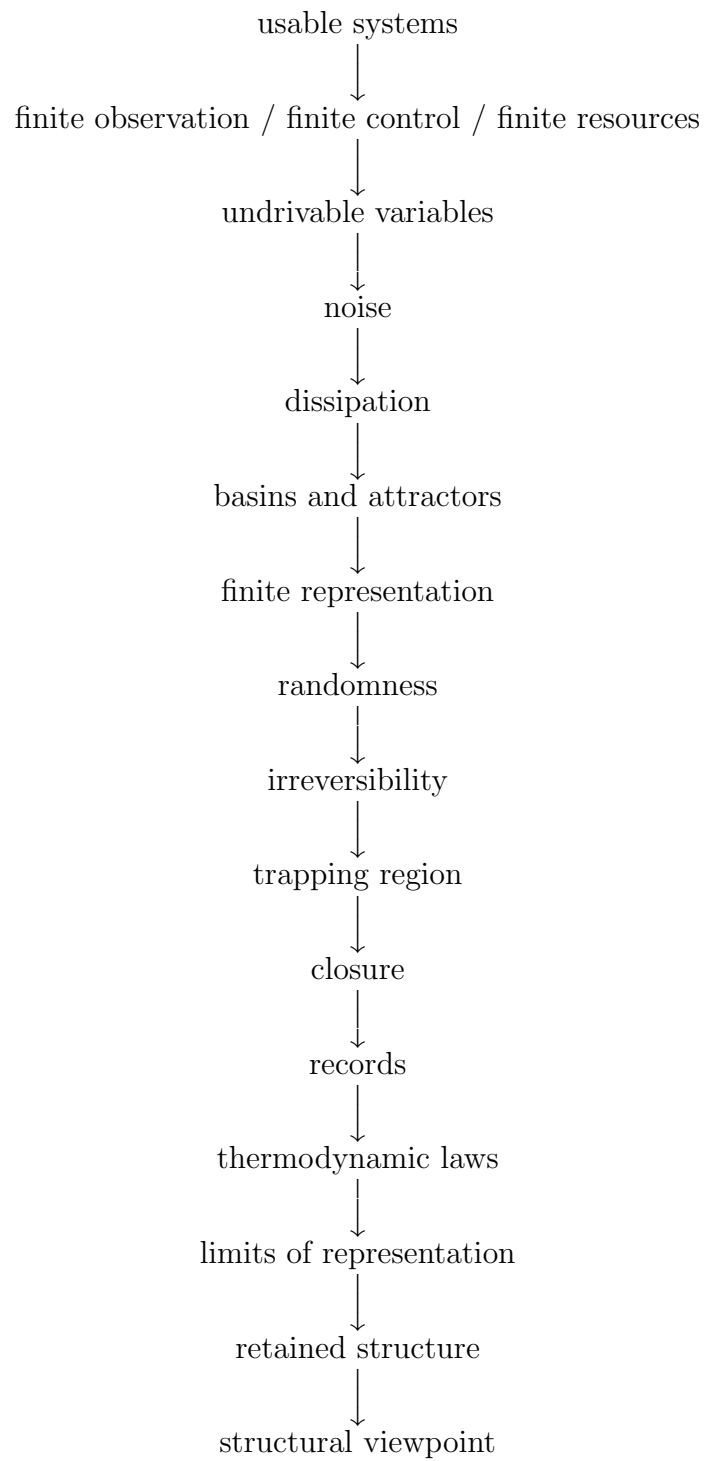
Its claims should therefore be read accordingly. What follows is not a complete theorem–proof treatment of nonlinear systems, but a structural framework that combines selective formalization with conceptual synthesis. The force of the argument lies in recurring constraints, recurring tradeoffs, and recurring structural consequences across domains.

Three themes will appear throughout the discussion: *usable systems*, *constraints*, and *boundaries*.

These themes are not introduced as formal definitions. They serve as guiding ideas for recognizing common structure across different contexts.

*The discussion that follows does not begin with solutions. It begins with limits.*

## Structural Roadmap



# Chapter 1

## Where the Problem Begins: Usability Before Exactness

Nonlinear systems are often introduced through their governing equations. From that perspective the central question appears to be whether the system can be solved: if the equations and initial conditions are known, the future evolution is determined.

Real systems are rarely encountered in this way. Observations are incomplete, measurements contain noise, and control inputs are limited. Some state variables may not be directly accessible at all. Under such conditions, the relevant question is no longer whether a trajectory exists, but whether the system remains usable.

This shift may appear subtle, but it changes how the problem is framed.

A trajectory may be understood simply as the path traced by the system state as it evolves over time. In the most basic form, this evolution may be written as

$$x_{t+1} = f(x_t)$$

where  $x_t$  denotes the state of the system at time  $t$ , and  $f(\cdot)$  describes the evolution rule. For nonlinear systems the function  $f$  is not restricted to linear form. In principle, once the initial state is specified, the trajectory is determined.

Usable systems, however, rarely have complete access to this state description.

### 1.1 Existence vs Usefulness

A nonlinear dynamical system may possess a mathematically well-defined solution while remaining unusable in practice. Exact trajectories may exist but remain inaccessible. Small perturbations in the initial state may lead to large deviations in observed outcomes. State variables may not be measurable with sufficient precision, and available inputs may not span the full state space.

Under such conditions the existence of a trajectory does not imply the possibility of reproducing it. Systems that remain usable therefore rely on representations that tolerate uncertainty. These representations do not attempt to preserve exact trajectories. Instead they preserve behaviors that remain stable under perturbations.

Mathematical existence and practical usefulness therefore describe different conditions.

### 1.2 Limits of Exact Dynamics

Exact dynamics describe how a system evolves given complete knowledge of its state and governing equations. Real systems rarely satisfy these assumptions. Initial conditions are known only within finite precision. Model parameters may drift, and external disturbances are unavoidable.

For many nonlinear systems, small uncertainties propagate through the dynamics. Prediction

errors accumulate over time. The governing equations may remain valid, but their predictive usefulness becomes limited.

Usable systems therefore rely on descriptions that remain stable under incomplete information. These descriptions emphasize invariant behavior, statistical regularities, or bounded responses rather than exact trajectories.

Linear models often remain usable because their behavior is structurally stable under perturbations. Nonlinear systems frequently lose this property, and the gap between exact dynamics and practical usability becomes more pronounced.

### 1.3 Usability as a Primary Constraint

Systems that operate under finite precision cannot prioritize exactness alone. Limited observation, limited control authority, and environmental disturbances introduce unavoidable deviations between models and physical systems.

Design under such conditions therefore emphasizes stability under perturbations. This requirement introduces structural compromises. Increasing sensitivity may improve short-term accuracy while reducing robustness. Increasing robustness may reduce achievable precision.

Such compromises appear across many domains, including control systems, statistical inference, and machine learning. They arise whenever systems must remain usable under finite resources.

### 1.4 Restatement

These observations suggest a shift in perspective. For real systems the primary condition is not exactness, but usability.

Exact dynamics remain relevant, but only within constraints imposed by finite observation, limited control, and environmental disturbances. A system must remain usable before exact descriptions become meaningful.

## Chapter 2

# Noise as the Boundary of Operational Systems

Noise is commonly introduced as random disturbance. In operational systems it often appears in a different role. Systems interact with variables that cannot be directly controlled or fully observed. These variables influence system behavior while remaining outside the domain of available inputs and measurements.

Under such conditions noise marks the boundary between the operational system and its environment. The distinction between signal and noise therefore reflects limits of control and observation rather than randomness alone.

### 2.1 Noise as a Structural Boundary

Operational systems are defined by the variables they can manipulate and observe. Variables outside this operational boundary cannot be directly driven, yet their influence still enters the system through disturbances, fluctuations, or unknown inputs.

From the perspective of the operational model these influences appear as noise. Noise therefore indicates the presence of degrees of freedom that remain outside the controllable structure of the system. In this sense noise is defined by structural position rather than statistical behavior.

Consider a system

$$x_{t+1} = f(x_t, u_t, w_t)$$

where  $x$  denotes the system state,  $u$  the controllable inputs, and  $w$  additional variables influencing the evolution.

Operational models often adopt a reduced description

$$x_{t+1} \approx \tilde{f}(x_t, u_t)$$

in which the influence of additional variables  $w$  is not explicitly represented.

Variables whose influence cannot be directly driven or incorporated into the operational description appear as noise. Noise therefore does not precede the model; it emerges from the boundary of the model itself. In this sense, noise first names unmodeled or operationally inaccessible influence; whether that influence is stochastic is a further question, not the starting definition.

### 2.2 Uncontrollable and Unobservable Subspaces

Control theory describes system dynamics through controllable and observable subspaces. The controllable subspace contains states reachable through available inputs. The observable subspace contains states that can be reconstructed from measurements.

Real systems rarely span the entire state space through these subspaces. Portions of the system remain unreachable or unobservable. Dynamics evolving in these regions cannot be directly manipulated or reconstructed, and their influence appears as noise in the operational description.

## 2.3 Internalization and State Expansion

Systems sometimes reduce apparent noise by expanding their internal representation of state variables. External disturbances may be incorporated into an enlarged state description, shifting certain sources of noise from the environment into the internal model.

State expansion reduces unexplained variation but increases system dimensionality. Additional states require additional measurements, parameters, and computational resources. The boundary between signal and noise therefore depends on the level of description adopted by the system.

## 2.4 Determinism and Randomness

Noise is often associated with stochastic behavior. However the distinction between unmodeled influence and stochastic process should be kept separate. The former concerns structural position relative to the operational model; the latter concerns a particular way of representing the unresolved dynamics. When viewed from the operational boundary of a system, the distinction between deterministic and random dynamics becomes ambiguous.

Variables that remain outside controllable and observable domains influence system evolution without becoming part of the internal model. From the perspective of the operational description their behavior appears unpredictable.

Whether the underlying dynamics of these variables are deterministic or random cannot be determined from the operational model alone. At the boundary of representation the two may become operationally indistinguishable. This indistinguishability does not arise from randomness itself, but from the limits of representation.

Noise therefore reflects limits of operational access rather than a confirmed statement about intrinsic randomness. Stochastic models may be useful ways of organizing such influence, but they do not exhaust what noise means in the present structural sense.

## 2.5 Why Noise Cannot Be Eliminated

Eliminating noise would require complete control and observation of all variables influencing the system. Physical systems rarely satisfy these conditions. Interactions with surrounding environments introduce variables that remain outside operational reach.

Operational models therefore manage noise rather than remove it. Filtering, estimation, and stochastic modeling reorganize the influence of undriven variables without eliminating them. Noise persists whenever systems interact with degrees of freedom that cannot be directly driven.

## 2.6 Restatement

Noise appears at the boundary of operational systems.

Variables that cannot be driven function as noise.

---

At this boundary deterministic and random dynamics become operationally indistinguishable.





## Chapter 3

# Dissipation as a Structural Condition

Dissipation is often treated as a secondary effect of physical implementation. In operational systems it occupies a different role. Systems that remain stable under perturbation generally do not preserve all distinctions among trajectories. Differences are compressed, forgotten, or absorbed over time. Under such conditions dissipation is not an external correction added to an otherwise complete structure. It is part of the condition under which robust structure appears at all.

Noise, as introduced in the previous chapter, represents unresolved interaction with variables outside the operational boundary. These interactions are not neutral. Their effects accumulate and reshape the evolution of the system. Dissipation reflects the manner in which such unresolved interactions are absorbed into the system as persistent structure.

### 3.1 Degeneracy in Non-Dissipative Systems

Consider a dynamical system

$$x_{t+1} = f(x_t)$$

where  $x$  denotes the state of the system. The set of all possible states forms the state space in which the evolution unfolds.

In a non-dissipative system distinctions between trajectories are preserved. Nearby states evolve without systematic loss of separation, and perturbations do not decay over time. Exact trajectories may remain well defined, yet the system retains every microscopic distinction among them.

This absence of contraction implies that no trajectory is eliminated. All possible evolutions coexist without selection.

For operational systems this creates a structural difficulty. Small differences persist indefinitely and must continue to be tracked. Exact evolution may still exist, but stable large-scale behavior cannot be maintained.

Without dissipation there is no mechanism for trajectory selection.

### 3.2 Basins and Structural Robustness

Operational systems must tolerate finite perturbations in state, input, and environment. This requires trajectories starting from nearby states to remain behaviorally related over time. Such behavior cannot arise from the preservation of distinctions. It requires their reduction.

Dissipative dynamics provide this mechanism by compressing regions of state space toward common outcomes. Distinct trajectories are not preserved; they are progressively merged.

Under these conditions basins of attraction appear. Different initial states evolve toward the same attractor, and perturbations within a basin do not destroy the resulting behavior. A basin

therefore represents a region in which trajectory differences are actively suppressed.

Robust structure does not arise from the preservation of trajectories, but from their convergence into stable domains.

A basin requires contraction in state space.

### 3.3 Limits of Linear Systems

Linear systems illustrate this limitation in a particularly clear form. Consider a linear system

$$x_{t+1} = Ax_t$$

where  $x_t$  denotes the state of the system at time  $t$ , and  $A$  is a linear transformation represented by a matrix.

In such systems the evolution preserves superposition. Trajectories do not interact in a state-dependent way, and no mechanism exists for selective contraction beyond uniform scaling. Distinctions between trajectories are not eliminated; they are merely transformed.

As a result linear systems do not generate the same kind of state-dependent structural differentiation that nonlinear dissipative systems do. They may exhibit uniform contraction, local stability, or attraction around an already given structure, but they do not by themselves produce the selective compression that this book treats as the formation of robust structure.

Structure therefore does not arise from linear dynamics. It requires interactions that modify the evolution of trajectories in a state-dependent manner.

Linear systems describe structure but do not generate it.

### 3.4 Dissipation as a Necessary Condition

Observable structure requires that different trajectories eventually become indistinguishable at the level of observation. Without such convergence every perturbation would lead to a permanently distinct trajectory, preventing the formation of stable patterns.

This requirement implies a compression of trajectories. In a dissipative description distances between trajectories contract along the evolution. A minimal expression of this condition may be written as

$$\|f(x) - f(y)\| < \|x - y\|.$$

The inequality does not specify a particular physical mechanism. It marks the loss of distinguishable structure during the evolution. When such contraction is absent, trajectories remain separated and no basin structure can emerge.

Structure is therefore not the accumulation of trajectories, but the result of their reduction.

Observable structure requires trajectory compression, and trajectory compression requires dissipation.

### 3.5 Destruction Before Structure

Dissipation removes distinctions before it produces recognizable structure. Trajectories that fail to remain compatible with the operational constraints are not preserved. Their differences are

progressively erased as the evolution proceeds.

This process is not neutral. It does not retain all possibilities and then organize them. It selectively eliminates trajectories that cannot persist under sustained interaction.

Structure therefore does not appear by adding order to a neutral background. It appears only after unstable alternatives have been continuously removed. What remains is not the total set of possible trajectories, but the restricted subset capable of surviving ongoing loss.

In this sense structure is not constructed through accumulation alone. It is revealed through elimination.

*Destruction is not the opposite of structure. It is a condition for its emergence.*

## 3.6 Dissipation and the Arrow of Time

A dissipative system distinguishes past from future through irreversible compression of state distinctions. As trajectories converge, differences that once separated them are no longer recoverable from the later state. This loss is ordered: distinctions are removed in the direction of evolution, but not restored. The system retains the result of compression while discarding the information required to reconstruct prior states.

Temporal direction therefore arises from the asymmetric loss of recoverable distinctions. Time does not appear here as an external parameter imposed on the system. It emerges as a consequence of irreversible evolution, and is inseparable from the process that generates structure itself.

From this point on, the explicit representation of time will no longer be required. The structure that remains does not depend on its parameterization in time.

*The arrow of time is the trace of dissipation.*

## 3.7 Restatement

Dissipation is not a secondary imperfection added to an otherwise complete system. It is a structural condition under which basin formation, robustness, and temporal direction become possible.

Without dissipation systems may remain exact while failing to remain stable. All distinctions are preserved, yet no structure can persist.

With dissipation systems lose recoverable detail, but stable structure begins to appear. What persists is not what is preserved, but what survives sustained elimination.

What appears as noise at one level becomes structured behavior at another. As interactions accumulate, unresolved dynamics do not disappear. They are reorganized into forms that can no longer be fully retained within finite descriptions.

*Structure is filtered through dissipation.*



## Chapter 4

# Randomness and Dimension

Noise described the boundary of controllability. Dissipation explained the emergence of structure. As interactions accumulate, unresolved dynamics are not eliminated. They are reorganized into forms that cannot be fully retained within finite descriptions.

Finite descriptions arise not only from physical limitations, but also from the need to balance usability and accuracy. Randomness appears when finite descriptions represent dynamics that exceed their capacity. Noise describes limitations of control, while randomness describes limitations of representation.

Randomness is often treated as a primitive feature of nature. In many systems it plays a different role. When deterministic dynamics are represented through finite descriptions, part of the underlying behavior cannot be retained. The unresolved portion of the dynamics then appears as randomness.

This chapter examines how randomness emerges as a consequence of finite representation.

### 4.1 Representation and Collapse

A finite description may be viewed as a mapping

$$\phi : X \rightarrow \hat{X}$$

from the underlying state space  $X$  to a representable space  $\hat{X}$ . Such a description cannot preserve all distinctions present in the underlying dynamics. Different trajectories may be mapped into the same representable state.

When multiple distinct states satisfy

$$\phi(x_1) = \phi(x_2), \quad x_1 \neq x_2,$$

the distinction between them is no longer retained.

Randomness appears when multiple underlying states are mapped to the same representable state.

The unresolved portion of the dynamics may be interpreted as residual dynamics. They do not arise independently, but as a consequence of representation collapse. This loss is not a property of the dynamics themselves, but of the representation. What cannot be retained is determined by the limits of the description.

### 4.2 Structural Limits of Finite Description

In practice, finite descriptions encounter structural limits:

These limits appear in multiple forms. They arise in observation, in control, and in computation. Each reflects the same constraint: not all dynamics can be retained.

**Observation.** Measurements aggregate underlying states into finite resolution. Distinct configurations may produce identical observations. The unresolved differences appear as variability.

**Control.** Control inputs act through limited channels. Not all degrees of freedom can be independently driven. Uncontrolled dynamics persist and influence outcomes in ways that cannot be fully specified.

**Computation.** Finite models operate with bounded precision and resources. State spaces are discretized, approximated, or truncated. Different trajectories may be treated as equivalent within the computation.

These are not separate phenomena. They are different manifestations of the same limitation: finite descriptions cannot retain all distinctions present in the underlying dynamics.

### 4.3 Accumulation and Reorganization

Unresolved dynamics do not disappear; they accumulate. As the system evolves, these unresolved components are repeatedly folded into the representation, and their effects are redistributed across observable behavior.

Over time, this accumulation produces variability that cannot be traced back to a single identifiable source within the model. The system exhibits fluctuations that reflect the aggregation of unresolved distinctions.

Rather than reconstructing their exact values, finite descriptions represent these unresolved components through their aggregate influence. In this sense, what remains of the unresolved dynamics appears as randomness within the model.

Randomness therefore does not arise from isolated events. It emerges from the accumulation and reorganization of unresolved dynamics.

### 4.4 Randomness as a Structural Consequence

Randomness is often interpreted as the absence of deterministic structure. In operational systems it reflects a different condition.

When a description cannot retain all distinctions, the unresolved differences appear as variability within the retained structure. This variability is not added from outside, but generated by the interaction between finite representation and underlying dynamics.

Once distinctions are no longer retained, their influence must still be represented. Randomness can therefore be interpreted as the trace left by representation collapse. It does not indicate that the underlying dynamics are random, but that the description no longer retains sufficient information to reproduce them.

Randomness is not defined by the absence of determinism, but by the limits of representation. Randomness records the cost of finite representation.

### 4.5 Determinism, Chaos, and Operational Equivalence

Deterministic dynamics can generate behavior that cannot be retained within finite descriptions. Sensitive dependence on initial conditions amplifies distinctions that cannot be resolved or tracked.

As a result, trajectories that are fully determined at the underlying level may become indistinguishable within the description. Their evolution appears unpredictable.

When this occurs, the distinction between deterministic and random descriptions loses operational meaning.

Randomness here does not indicate the absence of determinism. It reflects the inability of the description to retain the distinctions required to recover it.

Deterministic chaos is one instance of this condition. It demonstrates that apparent randomness can arise from fully deterministic dynamics when representation is limited.

## 4.6 Dimension as Representational Capacity

Dimension is often introduced as a property of a state space. In operational systems it reflects a different quantity. Here the term is used primarily in an operational sense: not only the geometry of the underlying state space, but the number of independent distinctions that a finite representation can retain.

Dimension describes the number of independent distinctions that can be retained within a representation. A finite description therefore imposes a bound on dimension. Not all degrees of freedom present in the underlying dynamics can be preserved.

When the number of independent distinctions in the dynamics exceeds the representational capacity, multiple states must be mapped into the same representation. This produces representation collapse.

Randomness appears when the effective dimension of the dynamics exceeds the dimension of the representation.

Dimension is therefore not only a geometric property. It is a measure of representational capacity. Finite systems operate under bounded dimension, and randomness reflects the overflow of structure beyond that bound.

A natural response to unresolved dynamics is to expand the description. If missing distinctions arise from discarded variables, including more variables may appear to restore determinism. In practice, this expansion encounters structural limits: observation, control, and computation remain finite, and not all degrees of freedom can be retained.

Increasing dimension may delay the loss of distinctions, but it cannot eliminate it. Finite descriptions must compress the dynamics they represent.

Randomness cannot be removed by increasing dimension. It is a consequence of finite representation.

## 4.7 Restatement

Finite descriptions cannot retain all distinctions present in the underlying dynamics. Multiple states are mapped into the same representation, and the unresolved differences accumulate over time.

Randomness does not arise from the absence of determinism. It arises from the limits of representation.

Structure records what survives compression. Randomness records the cost of collapse.





## Chapter 5

# Trapping Regions and the Limits of Reversibility

Previous chapters established several structural constraints of usable systems. Chapter 3 argued that observable structure requires trajectory compression. Chapter 4 showed that finite representation leaves residual dynamics that appear as randomness. The present chapter identifies the geometric condition that turns compression into observable structure.

In this chapter, observable does not refer to literal visibility, but to distinctions that can be retained and recognized within an operational description. Observable distinctions are those that remain distinguishable under the constraints of the representation.

The central claim of this chapter is precise. Reversible systems may contain mathematically exact structures, but without confinement they do not produce observable structure. A trajectory may exist, an invariant set may exist, and recurrence may occur, yet none of these by themselves create robustness. Observable structure begins only when dynamics can no longer escape a region of state space.

### 5.1 Reversible Dynamics and Empty Motion

Consider a discrete-time dynamical system

$$x_+ = f(x)$$

defined on a state space  $X$ . Here  $x_+$  denotes the successor state of  $x$ , without introducing an explicit time index.

**Definition 5.1.** The system is called *reversible* if the map  $f : X \rightarrow X$  is invertible.

If every state has a unique predecessor and a unique successor, then the dynamics can move indefinitely without losing information. The system evolves, but nothing is forgotten. Reversible systems can therefore exhibit perfectly exact motion without producing stable macroscopic structure.

The simplest example is a finite state space. If  $X$  is finite and  $f$  is bijective, then  $f$  is a permutation of the state space. The dynamics decompose into disjoint cycles, and every trajectory eventually repeats,

$$x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_k \rightarrow x_0.$$

The motion is exact, but it is empty in a structural sense. Nothing settles, nothing absorbs nearby trajectories, and nothing becomes stable under perturbation. A trajectory exists, but no finite region of states is drawn toward it.

This is the first important distinction of the chapter: trajectory existence is not the same as structural existence.

## 5.2 A Thought Experiment

It is useful to restate the same point without formal language.

Imagine a system moving forever inside a large state space. A special path may exist in that space. If the system starts exactly on that path, it may remain on it forever. In that sense exact structure may exist.

But this does not yet give observable structure. Observable distinctions are those that remain distinguishable under the constraints of the representation. Why? Because the system must start *exactly* on that path. If the initial condition is changed even slightly, the trajectory generally follows a different path. Nothing pulls nearby trajectories back.

For a structure to be observable, exact placement should not be necessary. A whole neighborhood of initial states should eventually display the same long-term behavior. If only one perfectly chosen initial state works, then the structure may be exact, but it is not robust, and therefore not observable in any operational sense.

This is the real limitation of reversible systems. They may contain exact trajectories, but they do not create a surrounding region that converges toward them.

Readers familiar with measure theory may interpret this as the statement that individual invariant trajectories typically occupy sets of measure zero in the surrounding state space. That formal language will be introduced next.

## 5.3 Measure Preservation and the Absence of Contraction

The previous intuition can be expressed in a single geometric condition: reversible systems preserve measure.

Let  $(X, \mu)$  be a measure space.

**Definition 5.2.** A dynamical system  $f : X \rightarrow X$  is called *measure preserving* if

$$\mu(f(A)) = \mu(A)$$

for every measurable set  $A \subset X$ .

This definition unifies both the discrete and continuous cases. For finite discrete systems,  $\mu$  may be taken as counting measure, so measure preservation reduces to preservation of cardinality. For Hamiltonian systems,  $\mu$  is phase-space volume, and measure preservation is precisely Liouville's theorem.

The meaning of this condition is direct. The dynamics may stretch, fold, or rearrange a set, but they do not compress its total measure. A cloud of initial conditions may be transported through state space without collapsing into a smaller set of positive measure.

This is why reversible systems fail to generate robust structure. Without contraction, nearby trajectories do not collapse into a shared future. They remain distinct, even if the motion itself is perfectly regular.

In finite discrete systems this appears as preservation of the number of states: a bijection cannot create a proper absorbing set by pulling in external states without sending an equal number back out. In continuous Hamiltonian systems the same logic appears as preservation of phase-space volume. The language is different, but the structural meaning is the same: no contraction, therefore no confinement.

## 5.4 Recurrence Without Attraction

A natural objection arises at this point. Measure-preserving systems often exhibit recurrence. Does recurrence not already produce structure?

The answer is no.

**Theorem 5.3** (Poincaré recurrence). *Let  $(X, \mu)$  be a finite measure space, and let  $f : X \rightarrow X$  be measure preserving. Then for almost every point  $x \in X$ , the trajectory of  $x$  returns arbitrarily close to its initial position infinitely often.*

This result describes an important feature of reversible motion, but it does not change the conclusion of the chapter. Recurrence means that a trajectory comes back near where it started, whereas attraction requires that nearby trajectories converge toward the same long-term behavior. These are fundamentally different notions.

$$\text{recurrence} \neq \text{attraction}.$$

A trajectory may revisit its neighborhood indefinitely without causing nearby trajectories to merge into a common future. Recurrence describes persistent motion. Structure requires confinement. For that reason Poincaré recurrence does not rescue reversibility from the charge of structural emptiness.

## 5.5 Basins and Observable Structure

We can now state more clearly what is missing.

To formalize this notion, we introduce a minimal definition.

**Definition 5.4.** Let  $A \subset X$  be an invariant set. The *basin of attraction* of  $A$  is the set of initial conditions whose trajectories converge toward  $A$ .

A basin is what turns mathematical existence into observable existence. Observable distinctions are those that remain distinguishable under the constraints of the representation. If a finite neighborhood of initial states leads to the same long-term behavior, then that behavior is robust under perturbation and can be observed without precise initialization. If no such neighborhood exists, then the structure may be exact, but it has no stability under finite uncertainty.

This is why no contraction means no observable structure. Without contraction, trajectories do not gather into common long-term behavior. Without such gathering, there is no basin. Without a basin, there is no robustness. And without robustness, there is no observable structure.

The logic is therefore

$$\text{no contraction} \implies \text{no basin} \implies \text{no robustness} \implies \text{no observable structure}.$$

At this stage we have not yet introduced the geometric condition that creates such basins. That condition is the next step.

## 5.6 Trapping Regions and the Emergence of Structure

The missing geometric condition is confinement.

A trapping region formalizes the idea that a portion of state space becomes dynamically closed under the evolution: once trajectories enter, they can no longer leave.

**Definition 5.5.** A measurable region  $R \subset X$  is called a *trapping region* if

$$f(R) \subset R$$

in discrete time, or equivalently if  $\phi_t(R) \subset R$  for all  $t \geq 0$  in continuous time.

Once a trajectory enters a trapping region, it cannot escape. The region may still contain complex internal motion, but the large-scale fact of confinement has already been established.

This is the crucial step from exact motion to observable structure. Observable distinctions are those that remain distinguishable under the constraints of the representation. A trapping region creates the possibility that many nearby trajectories remain together and approach the same long-term behavior. In that sense trapping is the geometric manifestation of trajectory compression.

Schematically,

$$\text{trapping region} \implies \text{basin} \implies \text{attractor} \implies \text{observable structure}.$$

This is why trapping regions belong at the center of the book's logic. They connect the dissipative argument of Chapter 3 to the observational argument of the present chapter, and they prepare the transition to records and irreversibility in the next chapter.

## 5.7 Reversible Systems Admit No Trapping Region

We can now state the first formal result.

**Proposition 5.6** (Reversible systems admit no trapping region). *Let  $f : X \rightarrow X$  be invertible and measure preserving. Then  $f$  admits no proper trapping region of positive measure.*

*Proof.* Assume that  $R \subset X$  is a trapping region of positive measure. This would mean that trajectories entering  $R$  remain confined within it under the dynamics.

Then

$$f(R) \subset R.$$

Because  $f$  is measure preserving,

$$\mu(f(R)) = \mu(R).$$

Thus  $f(R)$  is a measurable subset of  $R$  with the same measure as  $R$ .

No positive-measure contraction into a strictly smaller region can occur. A proper trapping region of positive measure cannot therefore be formed.  $\square$

This result clarifies the earlier intuition. Reversible systems may contain invariant sets, but they cannot generate the positive-measure confinement needed for observable structure.

## 5.8 Observable Structure Requires Trapping Regions

The converse structural principle can now be stated.

**Proposition 5.7** (Observable structure requires trapping regions). *Observable structure requires trapping regions.*

*Proof.* Observable structure must remain stable under finite perturbation. Observable distinctions are those that remain distinguishable under the constraints of the representation. Such stability requires that a finite neighborhood of initial conditions evolves toward the same long-term behavior, rather than dispersing.

This requires a basin of attraction of nonzero measure. A basin of nonzero measure requires that nearby trajectories, once sufficiently close, remain confined rather than spreading across the state space. Such confinement is precisely the role of a trapping region. Therefore observable structure requires trapping regions.  $\square$

This is the main structural result of the chapter. The issue is not whether exact invariant sets may exist. They may. The issue is whether such sets can become observable as stable structure under finite uncertainty. That requires trapping.

## 5.9 Exact Existence and Observable Meaning

We are now in a position to state the real takeaway of the chapter.

Reversible systems may contain exact structures. A periodic orbit may exist, an invariant torus may exist, and a special trajectory may persist indefinitely. But such objects generally occupy sets of measure zero in the surrounding state space and do not attract a finite neighborhood of initial conditions. They possess no basin, no robustness, and no observational stability.

For that reason they do not produce history or leave persistent macroscopic traces. They may exist mathematically, but they do not yet exist as observable structure.

Observable distinctions are those that remain distinguishable under the constraints of the representation. Observable structure begins only when dynamics cease to be freely exchangeable with the rest of the state space. Once trajectories cannot escape, a finite neighborhood collapses into a shared future. At that point structure becomes visible, robust, and meaningful.

This is also the bridge to the next chapter. A system can form records only when its trajectories are no longer fully reversible. Confinement is therefore not merely a geometric condition, but the precondition for memory, history, and irreversibility.

### Anchor statements

Exact structure may exist in reversible systems, but observable structure requires trapping regions.

Structure appears only where dynamics cannot escape.



## Chapter 6

# Information, Records, and Irreversibility

*The past exists only through records left by irreversible dynamics.*

Chapter 5 established the geometric condition for observable structure: trajectories must become confined within trapping regions. Once motion becomes confined, the system can no longer freely retrace all of its past trajectories. At that point, irreversible behavior appears.

The purpose of this chapter is to explain what irreversibility produces. The key structural chain is

$$\text{irreversibility} \rightarrow \text{records} \rightarrow \text{information} \rightarrow \text{history}.$$

Irreversible dynamics leave physical traces of past states. These traces are what we call records. Information is the interpretation of the correlations carried by those records. As records accumulate, they become history.

### 6.1 Irreversibility Creates Records

In a perfectly reversible system, every trajectory can be retraced exactly. Any physical trace created during the motion could, in principle, be undone by reversing the dynamics, and nothing persistent would remain. Irreversibility changes this situation. Once trajectories can no longer freely reverse, certain degrees of freedom retain correlations with earlier states.

**Definition 6.1.** A *record* is a physical correlation between the present state of a system and one or more past states.

Records are therefore physical objects: marks, correlations, or configurations that preserve traces of earlier events. Examples include measurement outcomes, memory states in a computer, and geological layers in rock. Irreversibility allows such correlations to persist. Without irreversible dynamics, records would continuously erase themselves.

### 6.2 Records Compress History

Records never store the full microscopic trajectory of a system. Instead, they preserve a reduced description of past events. A thermometer reading summarizes many molecular interactions. A digital log records only selected system states. A fossil preserves only a small subset of biological history.

Records therefore compress history. A small set of present variables represents a much larger set of past possibilities. Irreversibility is thus closely tied to historical compression: fine details of past trajectories are lost, while coarse summaries remain encoded in records.

## 6.3 Observation as Record Interpretation

As established in Chapter 5, observable distinctions are those that remain distinguishable under the constraints of a representation. Such distinctions require confinement and persistence within the system.

Observation does not create the structural chain introduced above; it operates on it. To observe a system is to access and interpret records that have been formed through irreversible dynamics. Measurements, sensor readings, and stored data are all physical records produced as part of this process.

The existence of information does not depend on the observer. Information is already present as correlations within the system's physical records. Observation merely reveals or interprets these correlations. In this sense, observation is a method of accessing information, not the fundamental source of it.

A record can therefore be understood as a confined distinction that persists, making it accessible to observation.

## 6.4 Maxwell's Demon and the Cost of Records

Maxwell's demon appears to challenge the second law of thermodynamics by using information about individual particles to reduce entropy. The paradox disappears when the role of records is considered.

To perform its task, the demon must obtain information about particles. Each such interaction produces a record. These records must be stored, updated, and eventually erased for the demon to continue operating. The limitation therefore lies in record management. Records cannot accumulate indefinitely without physical cost.

The demon fails not because energy extraction is impossible, but because maintaining and erasing records introduces irreversible processes. The issue is not simply energy expenditure; it is the bookkeeping required to preserve, update, and remove records under finite representational capacity.

## 6.5 Information, Entropy, and Control

In this chapter, information is used primarily in the sense of physically embodied correlation carried by records. Information theory provides a quantitative language for part of that structure. Entropy measures uncertainty about the microscopic state of a system, while information corresponds to correlations that reduce that uncertainty.

When records are formed, information becomes localized in certain degrees of freedom, while entropy typically increases elsewhere in the system. In this way information and entropy are not opposites, but complementary descriptions of how systems redistribute uncertainty through irreversible physical processes.

Control systems rely on the same mechanism. Feedback requires information about the system, such information is embodied in records, and those records guide control actions. Information theory, control theory, and thermodynamics therefore describe different aspects of the same structural constraint: information processing requires irreversible physical change.



## 6.6 From Records to History

Once records exist, they accumulate. A single record captures one moment, while many records together form a persistent description of the past. These accumulated records constitute what we call history.

Geological strata, biological evolution, and technological archives are all collections of such persistent records. Without irreversible dynamics, these records could not persist. Trajectories would simply reverse and erase their own traces. History therefore emerges from the accumulation of records produced by irreversible dynamics.

### **Anchor statement**

Irreversibility creates records.

Records encode information.

Accumulated records form history.



# Chapter 7

## Thermodynamics

*Thermodynamics does not begin with heat.*

*It begins with irreversible compression in state space.*

The previous chapter established that records require irreversibility. A record preserves part of the past while discarding the rest. Once histories are compressed into records, full reconstruction becomes impossible. Information is retained only in reduced form. The past survives as a compressed trace rather than a complete trajectory.

This chapter examines the structural mechanism that connects irreversibility to thermodynamic description. Irreversibility does not only destroy recoverability. It also eliminates trajectories. When distinct histories merge, the reachable portion of state space is compressed. Repeated compression confines motion to regions that trajectories cannot escape. These regions form the structural basis required for thermodynamic description.

The structural chain developed here is

irreversibility  $\rightarrow$  trajectory elimination  $\rightarrow$  trapping regions  $\rightarrow$  closure  
 $\rightarrow$  structure  $\rightarrow$  records  $\rightarrow$  thermodynamics.

Thermodynamics is therefore not the microscopic origin of irreversible systems. It is the description required once system histories have been compressed into closed regions of state space. In this sense, thermodynamics describes systems with compressed histories.

### 7.1 Irreversibility and Trajectory Elimination

Irreversibility eliminates trajectories.

In reversible dynamics, every state has a unique predecessor. The mapping between past and present states can be inverted, and individual trajectories remain distinguishable. Even when motion is complex, trajectories remain recoverable in principle.

Irreversible dynamics remove this property. Different histories can lead to the same present state. Once this occurs, the present no longer determines a unique past. Distinct trajectories collapse into a single observable continuation. Irreversibility therefore removes trajectories from the effective state description.

This change is structural rather than merely epistemic. When histories merge, the geometry of reachable behavior is altered. A system that repeatedly merges histories does not preserve the full set of possible trajectories. It progressively eliminates them.

### 7.2 Trajectory Compression

When trajectories are eliminated, the set of reachable states is compressed.

Conceptually this means that evolution preserves fewer distinctions than were present in the past. Distinct initial states are mapped into fewer distinguishable states. The effective set of

reachable states therefore shrinks under repeated application of the dynamics.

A minimal representation is given by a mapping

$$x^+ = f(x).$$

If the mapping  $f$  is not invertible on the represented state space, multiple states may satisfy

$$f(x_1) = f(x_2), \quad x_1 \neq x_2.$$

The future state therefore carries fewer distinguishable distinctions than the set of states that led to it. This is not merely a loss of information in a descriptive sense. It is a structural identification of previously distinct trajectories.

Dissipation, coarse representation, and measurement all produce this effect. In each case the system is not only evolving; it is reducing the number of distinguishable states. Trajectory compression is therefore the structural mechanism underlying dissipation, rather than a separate phenomenon.

### 7.3 Trapping Regions

Repeated trajectory compression confines motion to restricted domains.

A trapping region is a region of state space that trajectories may enter but cannot escape. Once motion has been compressed into such a domain, future evolution remains confined there. The region acts as a one-way container for reachable behavior. In this sense, the trapping region is the geometric form through which such a domain first appears.

In minimal form,

$$f(A) \subseteq A$$

where  $A$  denotes a subset of the state space that maps into itself under the dynamics.

This relation expresses the structural consequence of compression rather than serving as its definition. As trajectories are eliminated, escape routes are progressively removed. Once sufficient trajectories have been identified and collapsed, motion becomes confined to sets that map into themselves.

Trapping regions therefore arise as a consequence of irreversible compression. They are not additional assumptions imposed on the system, but the geometric result of trajectory elimination.

### 7.4 Closure and Structure

A trapping region becomes structurally meaningful when accumulated trajectories form a closure.

Trajectories confined to a non-escapable domain accumulate within a restricted set of possible histories. Their limit behavior remains within the same domain. The region therefore behaves as a closed invariant set under the effective dynamics.

Topologically, closure incorporates limit behavior into the trajectory set. Structurally, this removes the need to preserve individual paths. What remains relevant is the closed set generated by their accumulation.

The central structural statement of this chapter is

structure = closure of compressed system histories.

Structure is not the collection of original trajectories. It is the closed set produced after those trajectories have been irreversibly merged.

## 7.5 Records and Compressed History

Closed structures retain partial traces of the histories that formed them.

Once trajectories are compressed into a closed domain, the present state no longer determines a unique past. It reflects only the surviving effects of many merged histories. These surviving effects are records.

A record is therefore not a complete memory of what occurred. It is a compressed remainder carried by structure. Without persistent closed domains, compression would erase history completely. Closure allows partial traces of past trajectories to remain encoded in stable patterns of behavior.

Records are thus the historical aspect of closure. Structure stabilizes behavior in state space. Records stabilize the residual effects of how that behavior was formed.

## 7.6 Thermodynamics as Structural Description

Thermodynamics describes systems whose histories have been irreversibly compressed into closed domains.

When many microscopic histories merge, microscopic detail becomes inaccessible. A single observable thermodynamic structure corresponds to many compatible microscopic configurations. Such systems cannot be usefully described by reconstructing exact trajectories. They must instead be described in terms of compatible configurations.

As established in Chapter 5, observable distinctions are those that remain distinguishable under the constraints of representation. Chapter 6 then showed that observation operates by accessing records formed under those constraints. Here and below, *observable* should be read in that structural sense.

Entropy measures the multiplicity of microscopic configurations compatible with the same observable structure. In compact form,

$$S = \log W,$$

where  $W$  denotes the number of microscopic configurations consistent with the same observable description.

This relation expresses that observable structures correspond to many underlying possibilities. Thermodynamics therefore emerges when closed structure and compressed history coexist. It provides the statistical language required to describe such systems.

## 7.7 Thermodynamic Laws as Structural Constraints

Thermodynamic description requires surprisingly little.

Once system histories have been irreversibly compressed into closed domains, a direct description in terms of individual trajectories is no longer possible. To make such systems

describable beyond immediate dynamics, their behavior must be expressed in terms of relations between observable states.

Only two structural requirements are needed for such a thermodynamic description:

- Observable thermodynamic structures must be comparable.
- State transitions must admit consistent accounting.

From these minimal requirements, together with trajectory compression inside non-escapable domains, the familiar thermodynamic laws emerge.

### **Zeroth Law: Comparability**

Observable structures must be comparable.

Thermodynamic descriptions require that different systems can be placed in a relation that determines whether they are indistinguishable with respect to the same observable description. If two systems coexist without producing further changes, they belong to the same equivalence class.

The zeroth law establishes a consistent equivalence relation among observable states. It allows states to be compared, without yet introducing a numerical scale or ordering. Temperature arises only after additional structure is introduced. Structurally, the zeroth law guarantees that observable states form comparable classes.

### **First Law: Accounting Closure**

Observable structure transitions must admit consistent accounting.

When a system moves between observable structures, the changes must correspond to measurable exchanges with the environment, and the bookkeeping of those exchanges must close. In thermodynamics this appears as conservation of energy.

From the structural viewpoint, the first law ensures that transitions between compressed states remain describable through a consistent balance relation. Without such closure, observable transitions would not form a coherent description.

### **Second Law: Irreversible Compression**

Once histories are compressed, distinct microscopic trajectories become indistinguishable. Many microscopic histories correspond to the same observable structure.

As evolution proceeds, trajectory merging cannot decrease the number of compatible microscopic configurations. Entropy therefore increases because trajectory elimination hides microscopic distinctions.

In this framework the second law is not introduced as an independent axiom. It follows from irreversible compression inside non-escapable domains.

### **Third Law: Boundary of Distinguishability**

Finite representation imposes limits on how finely states can be distinguished.

As temperature approaches absolute zero, the number of distinguishable microscopic configurations approaches a bound. The third law expresses a boundary of distinguishability within the

compressed state description. It marks the limit beyond which further compression no longer produces new observable distinctions.

**Structural Implication** In this sense the thermodynamic laws do not precede the structural description developed here. They emerge from it.

Thermodynamics therefore describes systems after histories have already been compressed into observable records. It does not track individual trajectories. It counts the multiplicity compatible with the same observable structure.

At this point the thermodynamic argument of the chapter is complete. Irreversibility eliminates trajectories. Compression restricts reachable states. Closure stabilizes behavior into structure. Records preserve only residual traces of eliminated histories.

Thermodynamics is the language required for systems in exactly this condition. The thermodynamic argument requires no further extension at this stage. It persists here not as an added causal hypothesis, but as the residual consistency condition of compressed history. Its persistence should be understood not as an externally imposed theorem, but as the domain of description left once compressed histories can be compared only through retained states. Questions concerning the boundary of observation and the accessibility of such records belong to the next step of the book's argument.





## Chapter 8

# Limits of Representation: Outside the Boundary

*Not all structure can be retained within representation.*

*What cannot be retained does not disappear.*

*It appears as a boundary.*

The previous chapter argued that thermodynamics becomes necessary once system histories can no longer be preserved as individually recoverable trajectories. Irreversible compression confines motion, closure stabilizes behavior, and records retain only reduced traces of what occurred. At that point, structure no longer appears as exact evolution, but as what remains distinguishable after compression. Thermodynamics is the language of systems in that condition: systems whose microscopic histories can no longer be followed one by one, and must instead be described through the multiplicity compatible with the same observable record.

This chapter begins from the limit of that conclusion. If representation cannot retain all structure, then part of what shapes the system must remain outside the present closure of observation, recording, and description. This outside should not be understood as a metaphysical exterior or as a second world hidden behind the first. It is the structural remainder that cannot be kept within the available representation, yet continues to constrain what becomes observable, comparable, and stable.

In a mathematical sense, one may certainly speak of a domain determined by the assumptions under which a system is represented. From a functional-analytic point of view, such assumptions help define the admissible space within which states, operators, or trajectories are allowed to appear. That usage remains valid, and it is not rejected here. The present chapter, however, is concerned with a different question: not the admissible domain presupposed by representation, but the point at which representation can no longer retain all that shapes the system. Domain specifies where representation is defined. Boundary specifies where representation ceases to retain the distinctions required by structure. For that reason, the guiding term in what follows is not *domain* but *boundary*.

The central question of this chapter is therefore not what exists beyond structure in an absolute sense, but what must appear at the boundary once representation fails to preserve everything that participates in the dynamics.

The argument that follows is more speculative than the preceding chapters, but its method remains the same. Earlier chapters repeatedly discarded assumptions that could not survive finite observation, finite representation, and finite operation, and retained only what remained structurally unavoidable. The present chapter continues that movement. It does not attempt to derive modern physical theory from the framework developed here, nor does it replace the established content of relativity or quantum theory. Instead, it asks what kinds of boundary, branch, label, and residual structure become thinkable once representation, distinguishability, record formation, and closure are treated as primary constraints. What follows should therefore be read as a structural proposal: not a completed derivation, but a disciplined extension of the central logic developed in this book.

## 8.1 The Boundary of Observation

Observation is often described as though it were a window opened onto a system that already exists in complete form. From that perspective the main question appears to be whether the window is clear or blurred, broad or narrow, accurate or noisy. The previous chapters suggest a stronger conclusion. Observation does not begin with a complete world and then lose part of it. Observation begins only where distinctions can be retained as records under finite resolution, finite storage, and finite coordination. What cannot be retained in that way does not simply arrive as imperfectly seen content. It fails to enter the same structure at all.

This shift removes a familiar but misleading assumption. The boundary of observation is not merely the edge of an instrument or the temporary limit of current technique. It is a structural limit introduced by representation itself. A distinction counts as observable only if it can survive as a retained difference within the available record. If no such retention is possible, then the distinction does not belong to the same closure of observation, even if it continues to influence the dynamics. The relevant contrast is therefore not between the visible and the invisible in a naive sense, but between what can enter the same representational closure and what cannot.

In minimal form, one may write observation as a compression map

$$\phi : H \rightarrow R$$

from a space of possible histories  $H$  to a space of retained records  $R$ . If two histories satisfy

$$\phi(h_1) = \phi(h_2), \quad h_1 \neq h_2,$$

then the distinction between them is not retained within the present observational closure. Equivalently, one may define an indistinguishability relation

$$h_1 \sim h_2 \iff \phi(h_1) = \phi(h_2),$$

so that retained records may be regarded as equivalence classes induced by compression.

The outside introduced in this chapter begins at exactly this point. It is not a metaphysical exterior and not a second world hidden behind the first. It is the remainder of the dynamics that cannot be retained within the present closure of observation. Earlier chapters encountered this remainder in several forms. In Chapter 2 it appeared as noise at the boundary of control and observation. In Chapter 4 it appeared as unresolved distinctions under finite representation. In Chapter 7 it appeared again when many microscopic histories became inaccessible and only their multiplicity survived in the record. The present chapter removes the final simplification that these earlier discussions still permitted: the assumption that what lies beyond retention may be ignored once it has been excluded from the record.

It cannot be ignored, because exclusion does not remove influence. What remains outside the record continues to constrain what the record can be, what comparisons can be made, and what forms of structure can stabilize. Observation therefore has a boundary in a stronger sense than simple finitude. It is bounded not only because it sees less, but because what it cannot retain reappears as a condition on what can be retained. At that point the outside becomes structurally legible. It appears not yet as a positive inventory of hidden entities, but as boundary.

## 8.2 Finite Propagation and Observational Horizon

Once observation is understood in terms of record formation, a second assumption can be discarded. It is not enough that records exist. They must also be coordinated. A retained distinction that cannot be propagated, compared, or returned cannot function within the same observational closure as distinctions that can. The issue is therefore not simply whether a signal exists, but whether records can enter a common structure of comparison. Propagation limits are structural because they constrain not only access, but the possibility of shared retention.

From this point of view, finite propagation does more than slow communication. It divides the system into regions that can still participate in a common record and regions that cannot. A record that arrives too late, cannot return, or cannot be brought into comparison with other records does not merely reduce precision. It alters the closure of observation itself. Some parts of the dynamics remain within a common horizon of coordination. Others pass beyond it. What had first appeared as a practical limit on signaling must therefore be recognized as a structural limit on observational closure.

This may be expressed by introducing an accessibility relation  $\mathcal{A}(o)$  for a given observer or observational closure  $o$ . If a region or event  $r$  satisfies

$$r \in \mathcal{A}(o),$$

then records from  $r$  can still enter the same coordinated closure. If instead

$$r \notin \mathcal{A}(o),$$

then the corresponding records cannot be brought back into that closure in a form that restores comparability.

This is the sense in which horizons arise. A horizon is not defined here by a particular physical model, but by a failure of coordinated retention. Beyond a horizon, records may still exist locally, and dynamics may continue, but they no longer enter the same closure in which observation forms comparable structure. A horizon is therefore more than an observational inconvenience. It marks the point at which the outside becomes organized by finite propagation. Some regions remain inaccessible only relative to current means. Others are inaccessible in principle for a given closure because no record can return in a form that restores comparability.

Light speed enters this discussion as the simplest and most familiar form of this condition. If information cannot propagate arbitrarily fast, then no observer can gather all records into a single instantaneous closure. Black holes sharpen the point by presenting a limiting case in which return fails entirely across a horizon. The examples differ in scale and physical detail, but their structural meaning is the same. Finite propagation makes observational closure local, and whatever cannot be brought back into that closure appears at first not as a new object of knowledge, but as horizon.

## 8.3 General Relativity and the Geometry of Accessibility

Finite propagation removes another simplifying assumption. If records cannot all be coordinated within a single global closure, then geometry cannot remain a neutral stage on which dynamics merely happen to unfold. Geometry begins to matter structurally because accessibility itself becomes part of the problem. Which regions can be reached, which records can be compared, and which paths can return to a common closure are no longer secondary questions. They belong to the condition under which structure can be retained at all.

At this point the problem is no longer adequately stated in terms of signal delay alone. Delay already presupposes a geometry within which paths are fixed and comparison remains external to the distribution of structure. But once propagation, return, and coordination depend on the system's large-scale organization, accessibility must itself be represented. The geometry relevant to observation can no longer be treated as a passive background. It becomes part of the structural description because it governs the very possibility that records enter the same closure.

General relativity may be placed at exactly this threshold. Its significance here is not that it is reconstructed from the preceding chapters, nor that the present framework replaces its formal content. Its significance is that it provides a mature physical language for a condition that the structural argument has now made unavoidable: accessibility geometry must sometimes be treated as dynamically involved in what can be observed, compared, and retained. Once that happens, geometry is no longer outside the problem of structure. It enters the problem from within.

This provides a cleaner interpretation of why general relativity belongs in the present chapter. It does not appear here as a separate branch of physics imported for completeness. It appears because finite propagation and observational closure have already removed the assumptions under which geometry could remain external. If structure cannot be globally retained, and if records cannot be globally coordinated, then the geometry of accessibility becomes part of what the system is. General relativity is one way that this condition becomes explicit. In the logic of this book, it marks the point at which the outside is no longer merely what escapes representation, but what reshapes the paths by which representation becomes possible.

## 8.4 Distinguishability-Preserving Evolution

The preceding sections leave one further simplification in place. Even after observation has been bounded and accessibility has become geometric, one may still imagine that every stage of evolution must already belong to the logic of records. That assumption should also be discarded. Not all evolution begins by producing records, and not all structure begins by collapsing alternatives into a single retained outcome. Before record formation, there may be an interval in which distinctions remain present without yet becoming part of recorded history.

This interval is best understood in terms of distinguishability. A process may preserve alternative distinctions during evolution even when those distinctions have not yet been stabilized as records. In that case the system has not yet produced an observable outcome in the strong sense developed earlier in the book. It has instead preserved a structured field of possibilities whose internal differences continue to matter. The relevant question is therefore no longer whether the evolution is deterministic or indeterministic in the ordinary sense. The question is whether distinguishability is being preserved or compressed.

If distinguishability is represented by a relation  $D(\cdot, \cdot)$ , the contrast may be stated schematically. In a compression-dominated stage one expects

$$D(f(x), f(y)) < D(x, y),$$

whereas in a distinguishability-preserving stage the evolution satisfies, at least structurally,

$$D(f(x), f(y)) \approx D(x, y).$$

The point of these expressions is not metric precision, but structural contrast: one mode of evolution preserves distinctions, while the other eliminates them.

Wave behavior, coherent propagation, and certain forms of quantum-state evolution may be read from this point of view. Their structural interest lies not primarily in their formal differences from classical models, but in the fact that they suggest a mode of evolution in which alternatives remain distinguishable during propagation. Before record formation, the evolving state is not yet history in the sense established in Chapter 6. It does not yet preserve the past as a retained trace. It preserves a structured possibility whose distinctions remain unresolved.

This does not mean that record formation is absent from the system as a whole. It means that the relevant part of the evolution has not yet crossed the threshold at which distinguishability must be compressed into a retained outcome. The contrast is therefore not between reality and appearance, nor between wave and particle taken as primitive opposites. It is between distinguishability-preserving evolution and record-forming compression. Once that distinction is made, the path forward becomes clearer. One need not choose between continuity and discreteness at the level of the full system. One must instead ask where compression occurs, and what kind of record it produces.

## 8.5 Quantized Records and the Formation of Outcomes

Once distinguishability-preserving evolution has been separated from record formation, the role of quantization can be stated more precisely. Quantization enters this framework most clearly not at the beginning of evolution, but at the point where evolution becomes a record. A record cannot retain an unbounded continuum of distinctions. It must stabilize an outcome in a form that can be preserved, compared, and used within finite representation. Discreteness therefore appears here as a condition of record formation.

In minimal form, record formation may be written as

$$\psi : X \rightarrow R_f, \quad |R_f| < \infty,$$

where  $X$  denotes the pre-record state space and  $R_f$  the finite set of retained outcomes. Whatever richness the prior evolution may possess, the record itself belongs to a finite distinguishable class.

This point removes another misleading assumption. Quantization need not first be read as proof that the entire underlying evolution is discrete in the same way as the retained outcome. What is immediately justified is the more limited structural claim: measurement devices store records in finite distinguishable states, and observable outcomes inherit that finiteness. Even if the evolving system supports a richer continuity before record formation, the record itself cannot preserve all of it. Discreteness enters at the level of retained outcome.

From this perspective, measurement is the transition by which unresolved possibility becomes a record. What had remained distinguishable during propagation is forced into a retained form that no longer preserves the full structure of those alternatives. Quantization and measurement therefore belong together. Quantization names the discreteness of the retained record, while measurement names the compression through which that record is formed. The observable outcome is not the whole prior evolution. It is what remains after distinguishability has been reduced to a form that can persist.

Wave-particle duality can be read within the same structure. Wave behavior corresponds to distributed evolution before record closure. Particle behavior corresponds to localized record formation at the point where an outcome is retained. The apparent duality therefore does not require two separate ontologies. It reflects two structural conditions of the same system: one in which distinguishability is preserved across unresolved possibilities, and one in which distinguishability has been compressed into a discrete record. Once this is recognized,

quantization no longer appears as an isolated mystery. It appears as the point where finite representation forces structured possibility to become outcome.

## 8.6 Direction Without Time

The preceding sections have already displaced one familiar ordering principle after another. Observation no longer begins from total access. Propagation no longer permits a single global closure. Record formation no longer coincides with every stage of evolution. The deeper shift, however, is not introduced for the first time here. Chapter 3 already weakened the idea that time serves as the fundamental source of direction. There the arrow of time was read as the trace of dissipation: the result of irreversible compression rather than the primitive coordinate from which structure is derived. The present section begins from that result and asks a narrower but sharper question. If direction is not borrowed from an independent temporal axis, must the resulting direction be unique?

This question has already been prepared by the earlier chapters. Dissipation introduced a direction through irreversible loss. Records preserved the past only in reduced form. Thermodynamics described systems whose histories had already been compressed beyond exact recoverability. In each case, what mattered structurally was not the existence of an external parameter  $t$ , but the ordered elimination of distinguishable alternatives. What remained stable was not a point on an ideal time line. It was what survived compression.

Using the same compression criterion introduced above, directional order may be read without privileging time as an independent coordinate. The meaning of that criterion is structural. It says that ordering is present wherever distinguishable alternatives are systematically reduced. Time may still be used as a parameter of description, but it is no longer the source of direction. That point was already established earlier in the book. What matters here is the next step: direction is what remains when compression becomes ordered, and once it is understood in that way, its uniqueness can no longer be assumed in advance.

Once this shift is made, a new question becomes possible. If direction is defined by compression rather than imposed by an absolute temporal coordinate, must there be only one structurally meaningful direction available to the system? For linearized or already reduced descriptions the answer may appear obvious. For nonlinear systems it is less clear. A nonlinear system need not preserve a single globally uniform structure of reduction. Different branches of the same governing organization may support different compatible directions of compression. The issue is no longer whether time runs twice. The issue is whether the same system admits more than one structurally admissible orientation of ordered loss.

This does not yet yield a new ontology. It yields a stricter question. Once time is no longer taken as primitive, direction must be read from what the structure does rather than from the coordinate used to index it. The next step is therefore not to multiply temporal axes, but to ask whether symmetry can connect more than one admissible compression branch within the same governing system. If such a connection is required for structural completion, then branching is no longer a loose multiplicity of possibilities. It becomes a constrained relation between complementary orientations of the same governing structure. At that point the outside no longer appears only as horizon or inaccessible region. It begins to appear as branch.

## 8.7 Symmetric Branches and Antimatter

If direction is no longer fixed in advance by an absolute temporal coordinate, then symmetry also changes status. It can no longer be treated merely as an aesthetic regularity or as a secondary

property of equations already understood on other grounds. It may instead indicate that the same governing structure admits more than one compatible branch. A symmetry is important here not because it makes the system elegant, but because it prevents the retained description from being completed by only one orientation when the structure itself permits more.

In that setting, a branch is not just another solution written beside the first. It is a structurally admissible alternative within the same governing system. One may represent this schematically by a pair

$$\mathcal{B} = \{b_+, b_-\},$$

together with a symmetry relation

$$\sigma : b_+ \leftrightarrow b_-.$$

The symbols themselves carry little weight. Their role is only to mark that a single retained structure may admit complementary branches connected by symmetry without reducing one to noise or error.

**Proposition 8.1** (Symmetry and complementary branches). *If directional order is induced by compression rather than imposed by an absolute temporal axis, then a governing structure admitting symmetry completion may support more than one structurally admissible branch.*

This proposition does not specify the physical content of those branches. It states only that once direction is read from ordered compression, symmetry may require the retained structure to admit complementary orientations rather than a single privileged branch.

Antimatter may be approached from this point of view. Rather than introducing it first as a separate substance set opposite to matter, one may read it as a symmetry-related branch of the same structural system. The difference would then lie not in independent ontology, but in complementary orientation within the allowed branch structure of trajectories. This does not mean that the full content of particle theory has been derived from the present framework. It means only that antimatter becomes structurally thinkable once one allows that nonlinear systems may admit more than one compatible direction of compression under the same governing organization.

This reading also clarifies why antimatter belongs in the present chapter rather than in an isolated discussion of particle physics. The issue is not primarily which particles exist. The issue is how the outside becomes legible once retained structure can no longer be identified with a single branch of evolution. Earlier in the chapter, the outside appeared where representation failed to retain all distinctions. Here it appears again where structure cannot be completed by a single admissible orientation. A symmetry-related branch is therefore another way in which what lies beyond the present closure reenters the system: not as noise, not as horizon, but as a retained possibility that cannot be discarded without leaving the structure incomplete.

This remains a structural proposal rather than a completed deduction. Yet it performs an important purification of the question. If antimatter is introduced only as an additional object, the conceptual burden is shifted elsewhere. If it is introduced as a symmetry-related branch, the burden becomes local and structural. One must then ask what marks the difference between branches, how those marks are retained, and how interactions between branches become observable. At that point the chapter can move from branch to label, and from label to force.

## 8.8 Labels, Interactions, and Force

Once branching is admitted, a further simplification must be abandoned. It is no longer sufficient to say that multiple branches exist. If branches are to remain structurally meaningful, the system must retain some mark by which they remain distinguishable. Otherwise the branches collapse back into a single undifferentiated description. A branch that leaves no retained difference is not a branch within the represented structure. It is only an unexpressed possibility. Distinction therefore requires marking.

This is the point at which labels enter. A label is not introduced here as an arbitrary name attached from outside. It is the minimal retained mark by which structurally compatible branches remain distinguishable within the same governing system. If branch structure is represented schematically by  $\mathcal{B} = \{b_+, b_-\}$ , then one may associate a label map

$$\lambda : \mathcal{B} \rightarrow \Lambda$$

into a set of retained labels  $\Lambda$ , with

$$\lambda(b_+) \neq \lambda(b_-).$$

The content of this notation is simple. If the distinction between branches is to survive within the represented structure, it must survive as some retained difference.

Charge may be approached from this point of view. Rather than being introduced first as a primitive property carried by an already completed object, it may be read as a structural label that distinguishes complementary branches. This does not exhaust the content of charge in established theory, but it identifies a structural role that charge-like quantities can play within the present framework. They mark differences that cannot be compressed away without destroying the branch structure itself.

Once labels are retained, interaction can be reconsidered. If a system contains multiple distinguishable branches and labels, then its evolution can no longer be described only as the unfolding of isolated trajectories. Branches constrain one another through the way their retained differences enter the same structure. Interaction is therefore not best understood here as an external push applied between fully formed objects. It is better understood as structural reconfiguration under retained distinctions. Branches, labels, and closure constraints together determine which reorganizations remain admissible and which do not.

Force appears at the observable level of this reconfiguration. Within the present framework, force may be read as the form taken by structural adjustment when retained distinctions alter the accessible evolution of the system. It is not introduced as a primitive ingredient prior to structure. It appears when a nonlinear system can no longer evolve as though branch distinctions were irrelevant to its own closure. In that sense, force is the effective signature of structural reconfiguration. This is a structural reading, not a complete derivation of the forces recognized in established theory. Once this point is reached, a further distinction becomes necessary. Some reconfigurations remain local to branch and label structure. Others appear at the level of accessibility geometry itself. The latter belong to gravity.

## 8.9 Gravity as Geometric Bias

Gravity belongs to this chapter only if it is kept distinct from local interaction in the narrower sense. The previous section treated force as the observable form of structural reconfiguration among branches, labels, and closure constraints. Gravity points beyond that level. Its structural



role is broader because it appears not merely in the reconfiguration of retained distinctions within a fixed accessibility relation, but in the biasing of accessibility itself. The issue is no longer only how structure evolves within a geometry, but how geometry becomes non-neutral with respect to what is retained.

This is why gravity must be separated from force in the ordinary cataloguing sense. If accessibility were fixed, then all interaction could in principle be described against a common geometric background. But the earlier sections have already removed the assumptions needed for such neutrality. Propagation is finite. Horizons are real. Accessibility belongs to the structure rather than standing outside it. Under these conditions, some paths become privileged, deflected, or constrained at the level of the geometry itself. The system does not merely move through space. The accessible structure of motion is biased.

One may express this schematically by treating accessibility not as a fixed relation  $\mathcal{A}$ , but as one that depends on the retained structural distribution  $S$ :

$$\mathcal{A} = \mathcal{A}_S.$$

The notation is intentionally minimal. It states only that accessibility is no longer external to structure. The paths by which records propagate and return depend on what has already been retained as structure within the system.

Gravity may then be interpreted as the large-scale geometric bias induced by this dependence. It is not merely one force among others, because it does not first require a fixed stage on which forces act. It concerns the stage itself, insofar as accessibility geometry has become structurally involved in what can be coordinated, retained, and compared. This reading does not replace general relativity or claim to derive its formal laws from the present argument. It instead clarifies why gravitational behavior belongs on the geometric side of the boundary developed in this chapter rather than being grouped without remainder with local branch interaction.

This also explains why gravity belongs here, after the discussion of force rather than before it. Once interaction has been understood as structural reconfiguration, it becomes possible to see that not all reconfiguration is local to retained labels and branches. Some of it appears as a bias in the very paths by which records become mutually accessible. Gravity names this broader condition. It is the sign that the outside has entered not only through noise, horizon, branch, and label, but through the geometry of accessibility itself.

From this perspective, one may also understand why certain physical puzzles appear first as gravitational or geometric discrepancies rather than as ordinary local interactions. Dark matter, for example, may be read only as a tentative sign of the interpretive scope of this framework: not all retained structure need enter observation through the same channels. The point here is not to explain dark matter away, nor to identify it with undifferentiated residue, nor to open a new line of argument. It is only to note that structure may remain inaccessible to ordinary local record formation while still becoming legible through large-scale geometric bias.

## 8.10 Undifferentiated Residue and Structural Release

The preceding sections have progressively refined the outside into more definite forms. It first appeared as boundary, then as horizon, then as preserved distinguishability, then as branch, label, force, and geometric bias. Those forms describe ways in which what is not fully retained still reenters structure. Another possibility must now be considered. Not all remainder reenters as bias in accessibility or as retained branch difference. Some of it appears only when differentiated structure itself fails.

One final simplification nevertheless remains available, and it too should be discarded. When a structure can no longer retain its prior distinctions, the outcome is often described as though the relevant content had simply vanished. But loss of structural distinction is not identical to nothingness. The disappearance of a retained form does not imply the disappearance of everything that participated in that form.

This is the point at which undifferentiated residue becomes necessary. The term does not name a new substance hidden beneath all structure. It names what remains when prior distinctions are no longer retained as such. If a branch structure is released, if labels cease to be preserved, or if a previously stabilized organization can no longer maintain its differentiations, then what follows need not be interpreted as absence. It may instead be interpreted as the persistence of what is no longer retained in differentiated form. The residue is therefore undifferentiated not because it lacks all structure in an absolute sense, but because the structure by which it was previously tracked has been released.

Annihilation provides the clearest model for thinking about this transition. In the present framework, annihilation need not be read first as disappearance into pure nothing. It may be read as structural release: a case in which prior branch distinctions and labels are no longer retained, and the resulting state no longer supports the earlier differentiated description. Schematically, if a previously retained structure  $S_d$  loses the distinctions by which it had been stabilized, the transition may be written as

$$S_d \rightarrow R_u,$$

where  $R_u$  denotes undifferentiated residue. The expression is intentionally spare. Its purpose is only to distinguish structural release from simple negation.

If energy is mentioned in this context, it should be mentioned cautiously. The present argument does not yet justify a full redefinition of energy, nor does it require one. At most, energy may be approached here as a language for what remains available when differentiated structure has been released. Such a reading would fit the broader logic of the chapter, but it should remain subordinate to the clearer structural claim: what ceases to be retained as branch, label, or record need not cease to exist altogether. It may persist as residue after structural release.

## 8.11 Boundary Remark

The purpose of this chapter is not to complete the theory by naming everything that lies beyond the present boundary. It is to clarify how the boundary itself becomes legible once representation is taken seriously. The argument began by discarding the assumption that observation opens directly onto a complete structure. It then discarded the assumption that all records can be coordinated within a single closure, the assumption that all evolution already belongs to record formation, the assumption that time must serve as the primitive source of direction, and the assumption that interaction can be understood without reference to branch, label, or accessibility geometry. At each step, what could be discarded was removed. What could not be discarded reappeared as structure.

Seen in that order, the chapter has not assembled a catalogue of physical topics. It has followed a single line of reduction. What cannot be retained within representation does not thereby become irrelevant. It returns as boundary where retention fails, as horizon where coordination fails, as preserved distinguishability before compression, as branch where one orientation is insufficient, as label where branch difference must be retained, as force where retained differences reorganize evolution, as gravity where accessibility itself becomes biased,

and as residue where differentiated structure is released. The outside becomes thinkable only through these returns.

These should not be understood as final ontological categories. They are structural forms in which the outside becomes legible from within a system that cannot retain all of its own conditions of possibility. If the earlier chapters showed why usable systems cannot preserve complete history, the present chapter shows why what is not preserved does not simply disappear from the logic of structure. It reenters that logic as boundary. The next chapter can therefore return to synthesis from a sharper position. Structure is not complete in itself. It is what remains, together with the marks left by what could not be retained.



# Chapter 9

## Structural Viewpoint

*What survives is not what was, but what remains distinguishable.*

### 9.1 Structural Universality

Different scientific disciplines appear to study fundamentally different systems. Fluid dynamics studies flows, machine learning studies data, control theory studies feedback, and thermodynamics studies macroscopic physical processes. At the level of objects, terminology, and mathematical tools, these fields seem unrelated.

Yet when these systems are observed in practice, they repeatedly exhibit the same phenomena: noise, dissipation, randomness, and irreversibility. These are often treated as imperfections—limitations of instruments, incomplete information, or approximations in modeling.

The structural viewpoint developed in this book suggests a different interpretation. These phenomena do not arise from the failure of theory, but from the conditions under which systems become usable. Any usable system must be observed, represented, and acted upon through finite means. Observation has limited resolution, representation requires finite storage, and operation requires finite resources.

Under these constraints, complete system histories cannot be preserved. Trajectories are necessarily compressed. Compressed histories produce records, and accumulated records give rise to observable structure.

The similarities across scientific disciplines therefore do not reflect shared models or methods. They reflect a shared condition: all of these fields study systems that operate under finite constraints and are therefore structured through compression. The viewpoint summarized in this chapter concerns the retained domain of usable structure. It does not exhaust what conditions that domain from its boundary.

### 9.2 Usable and Operational Systems

The systems considered in this book are not arbitrary mathematical constructions. They are systems that can be interacted with in the physical world. Such systems must be observed, represented, and acted upon using finite resources.

Throughout this book, the term *usable system* refers to this class of systems. The term *operational system* emphasizes the same condition from the perspective of interaction: a system that admits consistent measurement, representation, and action. In this sense, all operational systems are usable systems, while the notion of usability extends beyond what is directly accessible through specific operations.

### Definition (Usable System)

A usable system is a system that can be observed, represented, and acted upon through finite resources.

This definition introduces three structural constraints. Observation is finite because measurement cannot resolve arbitrarily small distinctions. Representation is finite because storage and computation are bounded. Operation is finite because any intervention requires limited energy, precision, and coordination.

Systems that violate these conditions may exist as abstract mathematical objects, but they cannot function as usable systems. The structural phenomena discussed in this book therefore do not apply to all conceivable dynamical systems, but to the class of systems that remain stable under finite constraints.

A usable system is not one that preserves complete information. It is one that remains stable under loss.

Irreversibility removes exact histories. Compression reduces distinguishability. Closure stabilizes behavior. Records preserve only residual traces.

A usable system is therefore a system that survives by giving up exactness. What lies beyond that retained survival is not denied, but it is not the primary subject of this synthesis.

## 9.3 Structural Impossibility and Compromise

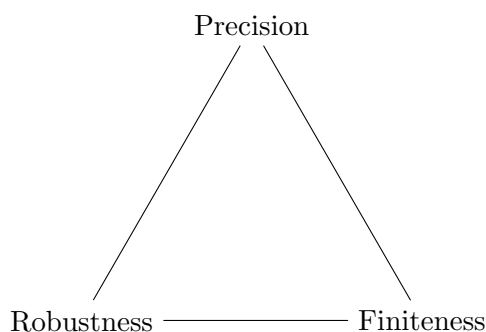
When analyzing usable systems, three objectives naturally arise. One may attempt to preserve exact trajectories, maintain stability under perturbations, and operate within finite observational and computational limits. These objectives, however, cannot be satisfied simultaneously.

- **Precision** refers to the preservation of exact trajectories and reversible dynamics.
- **Robustness** refers to stability under perturbations, typically associated with basins and attractors.
- **Finiteness** refers to operation under finite observation, finite representation, and finite resources.

These three conditions define a structural constraint rather than independent design goals.

### Structural Impossibility Principle

For systems operating under finite observation, finite representation, and finite operation, precision, robustness, and finiteness cannot simultaneously hold.



This triangle is not a matter of design preference. It is a structural condition for usable systems. Any such system must occupy a position within this space rather than satisfying all three objectives simultaneously.

Real systems resolve this constraint in a consistent way. They preserve robustness and finiteness while sacrificing perfect precision. Robustness ensures that behavior remains stable under perturbations, and finiteness ensures that the system can be observed, represented, and acted upon. Exact reconstruction of trajectories is therefore lost.

This loss of precision does not appear as a single effect. It manifests as approximate dynamics, stochastic behavior, noise, dissipation, and irreversibility. These are not imperfections of implementation or limitations of theory. They are structural consequences of systems operating under finite constraints.

A usable system is therefore not an ideal system degraded by error. It is a system shaped by unavoidable compromise. This compromise is a structural condition within the retained domain of usable systems, not a total theorem about all possible structure.

## 9.4 The Infinite-Dimensional Boundary

A natural question arises: could the structural impossibility described above be avoided if systems possessed infinite precision or infinite dimensionality? If trajectories could be preserved exactly, perhaps precision and robustness could coexist.

At first glance, this appears to offer a way out. If distinctions could be made arbitrarily fine, compression might seem unnecessary, and information loss might be avoided.

In practice, however, this question has little operational meaning. Before the state space of a system is fully explored, unknown degrees of freedom always remain possible. From the perspective of observation, unknown dimensions are indistinguishable from infinite dimensions.

One may therefore attempt to resolve the question by fully exhausting the state space. In that case, no further distinctions remain. The system becomes closed, and all future trajectories are determined by its structure. Evolution reduces to the unfolding of a fixed sequence.

This leads to two possibilities. Before the state space is exhausted, the question cannot be answered. After the state space is exhausted, the question no longer has meaning.

The issue is therefore not whether the world is finite or infinite in dimension. It is whether a system can be described, observed, and operated upon in a finite, usable way. A system that requires infinite precision cannot be used. If all distinctions were preserved, no compression would occur. Without compression, no records would form, and no stable description would exist.

The infinite-dimensional limit therefore marks a boundary of description rather than a realizable regime of operation. The issue is not only what lies beyond finite operation, but what cannot be retained within it.

## 9.5 Observation and Structural Accessibility

The structural constraints developed so far limit what systems can preserve. A further question then arises, though it can now be answered more briefly than in the preceding chapter: how does structure become accessible at all?

Observation does not introduce new structure. It defines the retained domain within which structure becomes accessible, distinguishable, and usable. Any usable system must be observed,

represented, and acted upon through finite means, and only those distinctions that survive such constraints enter the same operational closure as structure.

Accessible structure is therefore not the full set of system trajectories. It is the closure of compressed histories under finite observation.

Conceptually, the structural relation developed throughout this book can be expressed as

$$\text{Structure} = \text{closure}(\text{Compression}(\text{Histories})).$$

This relation is not a formal theorem, but a structural identity. Structure is not imposed on a system. It is what remains after histories have been compressed and only those distinctions compatible with observation are retained. The role of observation here is therefore not to reopen the full boundary question, but to specify the closure within which the present synthesis applies.

## 9.6 Four Structural Constraints

Once observation defines the boundary of accessible structure, the behavior of usable systems can be expressed in a minimal form. The preceding discussion does not introduce independent principles, but constrains how retained structure must arise within the usable domain.

These constraints can be compressed into four relations.

$$\begin{aligned} \textbf{Compression:} \quad & |\mathcal{T}_{future}| < |\mathcal{T}_{past}| \\ \textbf{Irreversibility:} \quad & f^{-1} \text{ not unique} \\ \textbf{Closure:} \quad & f(A) \subseteq A \\ \textbf{Records:} \quad & S = \log W \end{aligned}$$

The first relation states that under finite observation, distinguishable futures are fewer than the histories that produced them. The second states that once histories are compressed, unique reconstruction is lost. The third states that repeated compression stabilizes behavior into non-escapable domains. The fourth states that when many histories correspond to the same observable state, their multiplicity becomes measurable.

These relations do not extend the theory with additional laws. They summarize the structural consequences of usable systems under finite constraints.

## 9.7 Thermodynamics as Structural Language

The relations above do not merely describe isolated effects. Taken together, they locate the regime in which thermodynamics becomes the stable language of description.

Within the retained domain of compressed history, systems can no longer be described by tracking exact trajectories one by one. What remains available is the multiplicity of histories compatible with the same retained state, together with the relations by which such states can be compared and accounted for.

Thermodynamics arises as the language of this regime. It does not replace dynamics, but describes systems once compressed history has become the relevant descriptive condition.

The multiplicity relation stated above does not introduce a new principle. It expresses the consequence of compression: a single retained state corresponds to many underlying histories, and their number becomes part of the relevant description of the system.



In this sense, thermodynamics is not an external theory imposed on physical systems. It is the language of structure that remains once histories have been irreversibly compressed into comparable states.

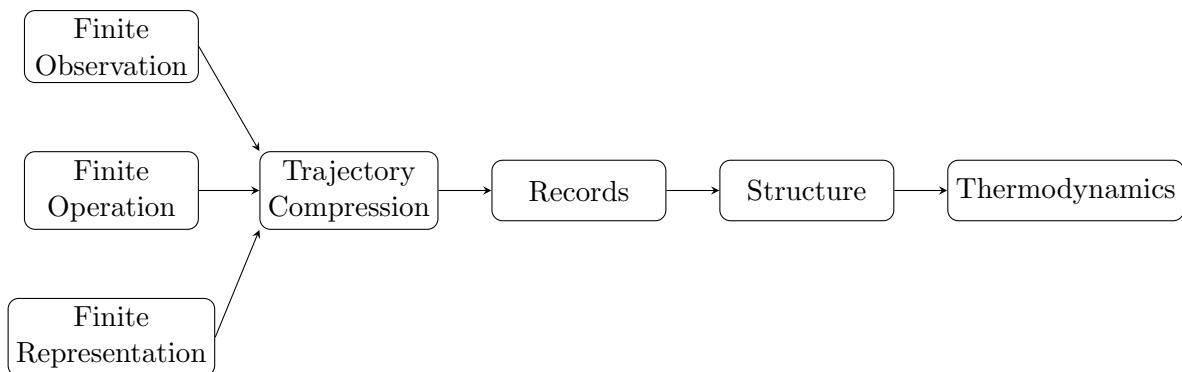
## 9.8 Structural Compass

The development of this book can be summarized as a directional relation.

Finite observation, representation, and operation prevent the preservation of complete histories. Histories are therefore compressed. Compressed histories produce records, records accumulate into structure, and thermodynamics provides the language used to describe such structures. The compass given below maps this development within closure. It is a diagram of usable structure in the retained domain, not an exhaustive map of everything that conditions that domain from its boundary.

Structure is the closure of compressed histories.

Thermodynamics is the language of such structures.



This diagram is not a complete description of systems. It serves as a compass. It indicates the direction in which usable systems evolve under finite constraints, and how structure emerges as a consequence of compression.

## Closing Remark

Nonlinear systems are not complicated versions of linear systems. They are systems that remain consistent under finite observation, finite representation, and finite operation.

When histories cannot be preserved, they are compressed. When histories are compressed, they become records. When records accumulate, they form structure.

What appears as noise, randomness, and irreversibility is not a failure of description. It is the condition under which structure becomes possible.

*What cannot be retained no longer appears as structure. What remains becomes structure.*



# Epilogue

Nonlinear systems are often described as systems that are too complex to understand. Their irregular behavior, sensitivity to perturbations, and apparent unpredictability are frequently taken as indications that the systems themselves are intrinsically complicated.

The perspective developed in this book suggests a different reading. Nonlinearity is not a consequence of excessive complexity, but of constraint.

Real systems do not evolve in an unconstrained space of possibilities. They operate within limits that restrict what can be distinguished, what can be retained, and what can be controlled. Under such conditions, histories cannot be preserved in full detail. They are necessarily reduced, and in that reduction distinct trajectories may become indistinguishable. What appears as unpredictability is not the failure of description, but the result of compression.

Within this perspective, several familiar notions take on a different meaning. Noise, randomness, and irreversibility are often treated as imperfections of measurement or limitations of theory. Here they instead appear as structural features of systems that cannot maintain complete histories. What cannot be distinguished is not retained, what is not retained does not simply disappear from the problem, and what remains accumulates into structure.

In this sense, nonlinearity is not an exception that complicates otherwise simple systems. It is the condition under which usable systems exist.

Seen from this viewpoint, the role of this book is necessarily limited. It does not provide a universal method for solving nonlinear systems. Instead, it clarifies the structural boundaries within which such systems can operate. Understanding these boundaries does not remove uncertainty, but explains why it cannot be eliminated. It does not remove irreversibility, but shows why it inevitably appears when histories cannot be fully preserved.

The framework developed here does not replace existing physical laws or mathematical models. It offers a structural perspective from which they can be interpreted, rather than extended or unified.

Recognizing structural closure does not imply that systems become limited or that meaningful behavior disappears. On the contrary, closure is what allows structure to exist. Without closure, trajectories would disperse without trace, histories would not stabilize, and records would not accumulate. Structure would not exist. Yet closure does not exhaust everything that conditions structure. It defines the retained domain within which structure can stabilize, while the boundary of that domain still shapes what can be distinguished, preserved, and compared.

The constraints discussed throughout this book are therefore not only limitations. They are also the conditions that make usable systems possible.

Once these boundaries are understood, a different question naturally arises. If structure inevitably forms under such conditions, how can it be recognized, organized, and used? That question lies beyond the scope of this work.

Nonlinearity is not an exception. It is the condition under which usable systems exist, and under which structure becomes possible.

*Although the state space may be closed, the structure ahead is not.*